WEEK 5 REPORT

Healthcare Data Cleaning & Data Modeling

(Power BI Desktop)

Submitted by

Preethi Pal D

Roll No: E25DA046

Department name: Bsc Data Science And AI

1. Introduction

This week I worked on a healthcare dataset instead of the stock data. The raw data had patient details, admission details and billing details all mixed in one big table, so my job was to clean it up and split it into a proper data model before it can be used for reporting.

2. Objectives

- Import and inspect patient records, admission types, and billing fields in Power Query.
- Handle null values in diagnostic codes and standardize categorical text attributes.
- Create derived conditional columns, for example Age Group buckets: 0-18, 19-35, 36-60, 60+.
- Structure dimension tables (Dim_Patient, Dim_Admission) and a fact table (Fact_Billing).

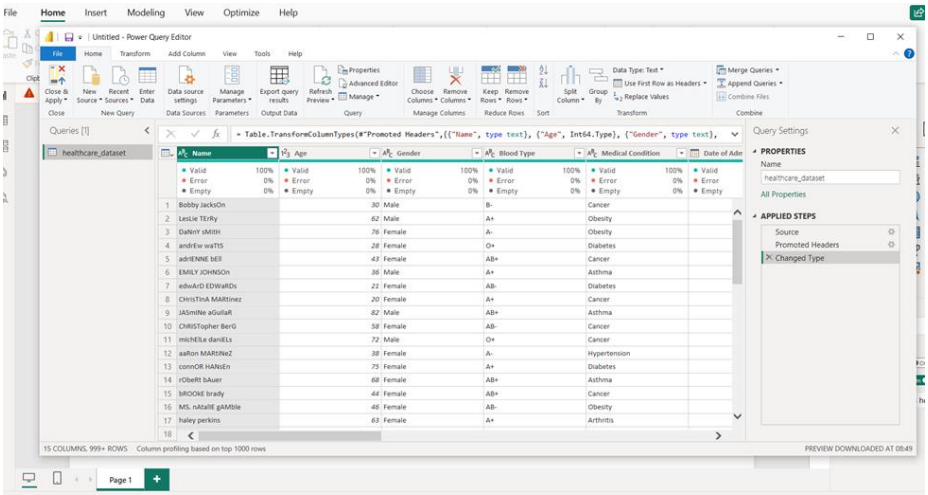
3. About the Dataset

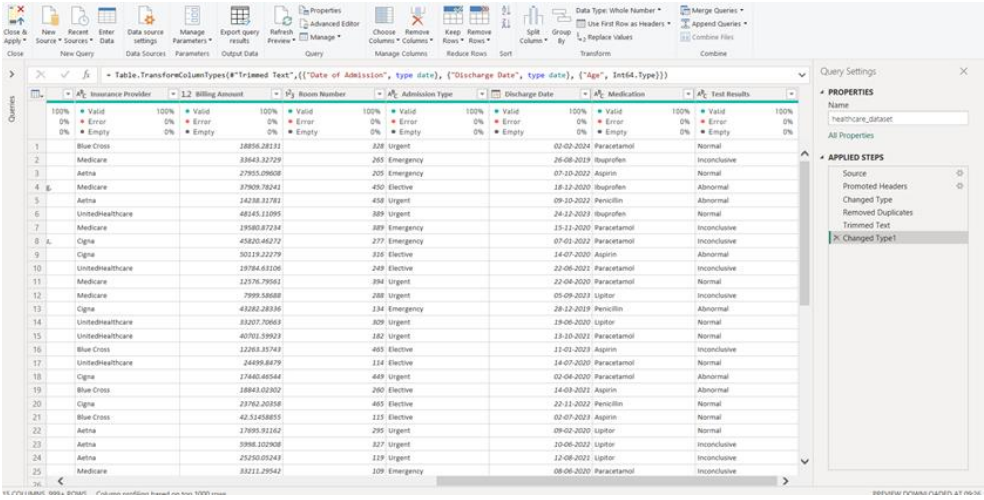
The dataset is called healthcare_dataset, and each row is one patient admission record. It has columns like Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, Hospital, Insurance Provider, Billing Amount, Room Number, Admission Type, Discharge Date, Medication and Test Results.

4. Steps

Step 1: Import and inspect the data

I loaded the healthcare_dataset file into Power Query. The first row was used as headers, and Power Query automatically set the data types for each column, for example Age as a whole number and Date of Admission as a date. All the columns showed 100% valid values with no errors.





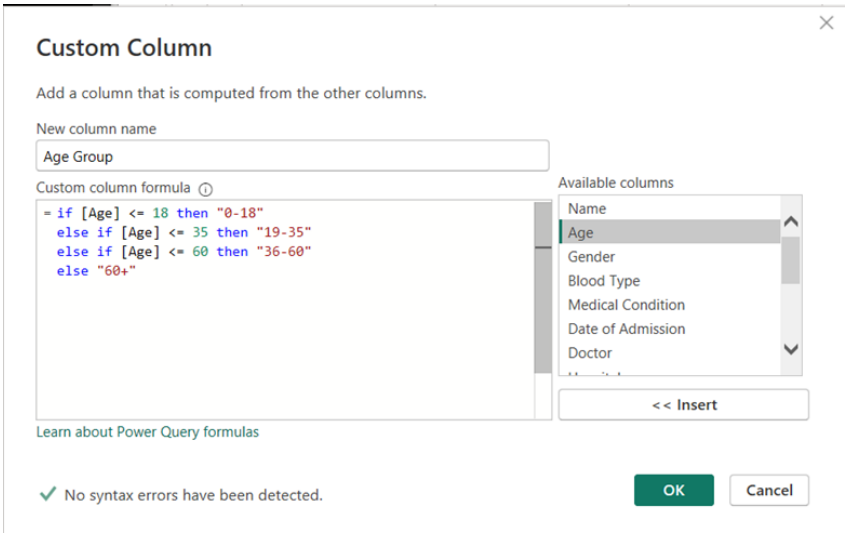
Step 2: Clean the data

Under Applied Steps I added Trimmed Text to remove extra spaces from text columns, and Removed Duplicates so that no patient record is repeated. I also fixed the data type of the Date of Admission, Discharge Date and Age columns so that dates are read as dates and Age as a whole number.

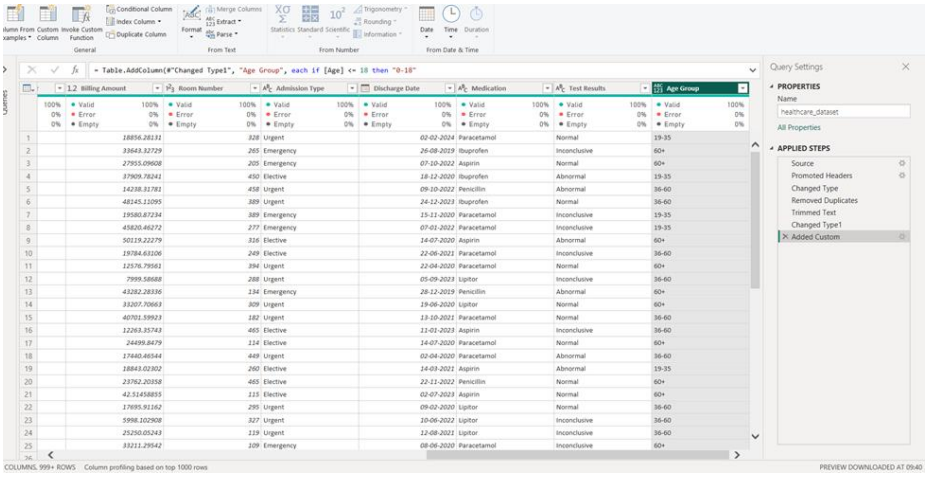
Step 3: Create the Age Group column

To group patients into age categories, I added a Custom Column called Age Group using this formula:

```
if [Age] <= 18 then "0-18"
else if [Age] <= 35 then "19-35"
else if [Age] <= 60 then "36-60"
else "60+"
```

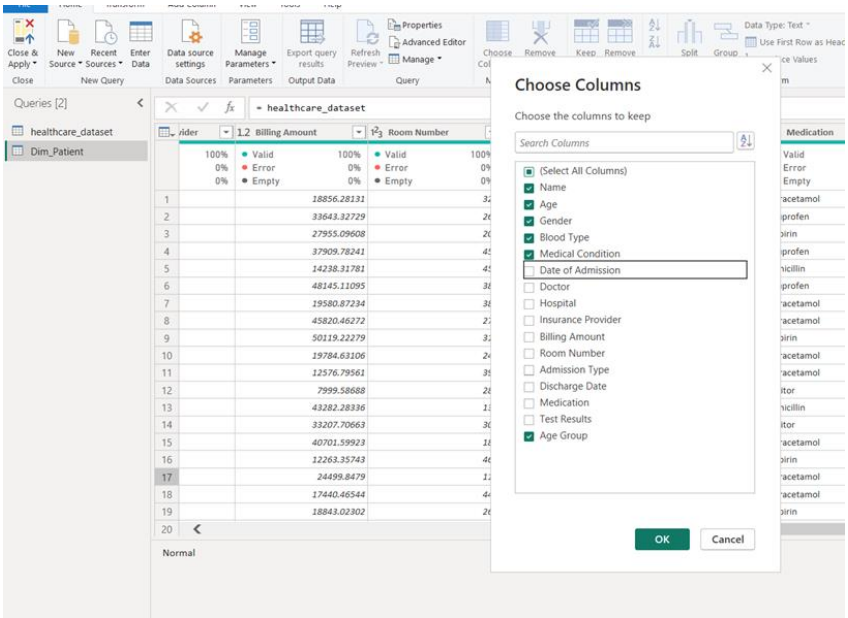


After clicking OK, the new Age Group column showed the correct bucket for every patient, for example 19-35 or 60+.



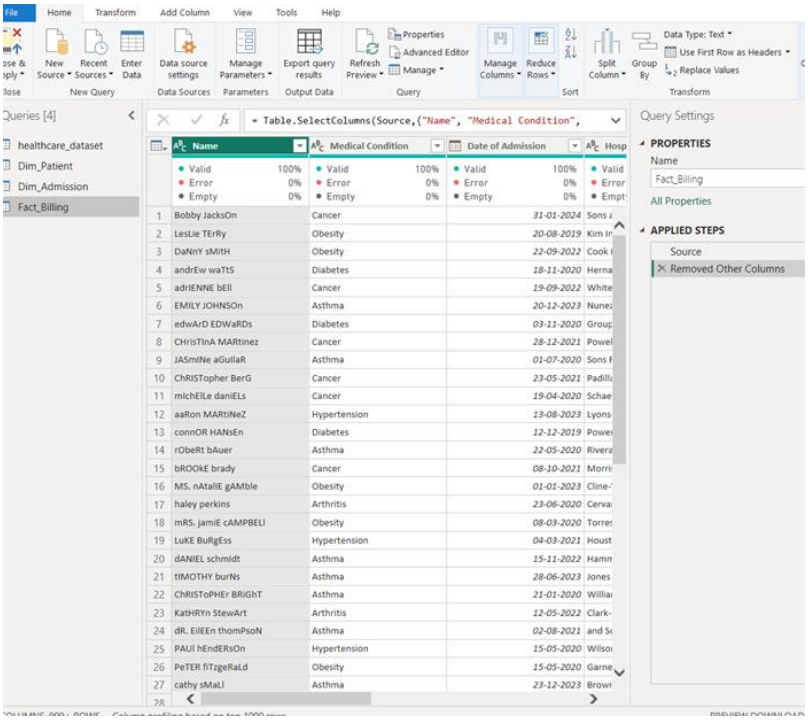
Step 4: Build the dimension and fact tables

The raw table had patient details, admission details and billing details all mixed together, so I split it into separate tables. I used Choose Columns to keep only the patient related fields (Name, Age, Gender, Blood Type, Medical Condition, Age Group) for a table called Dim_Patient.



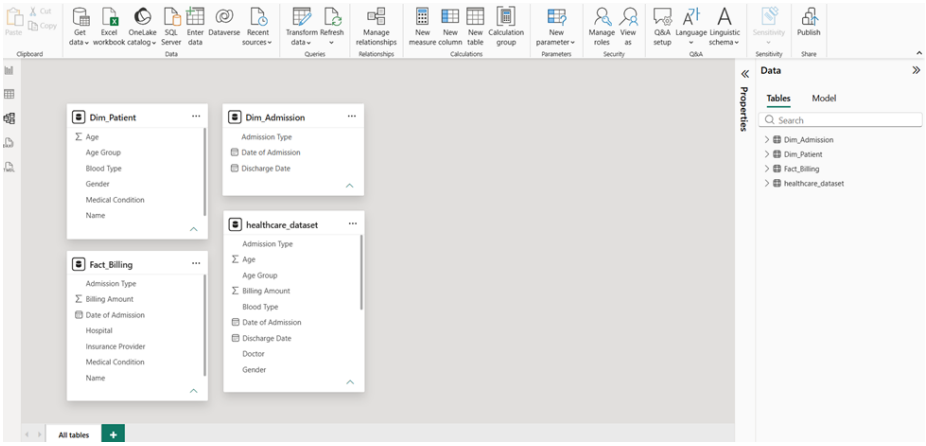
In the same way, I created a **Dim_Admission** table by removing the unnecessary columns and keeping only the admission-related fields such as **Admission ID, Date of Admission, Admission Type, and Discharge Date**.

And a Fact_Billing table by removing the columns that were not needed, keeping only the billing and admission related fields like Admission ID, Billing Amount, Date of Admission, Hospital and Insurance Provider.



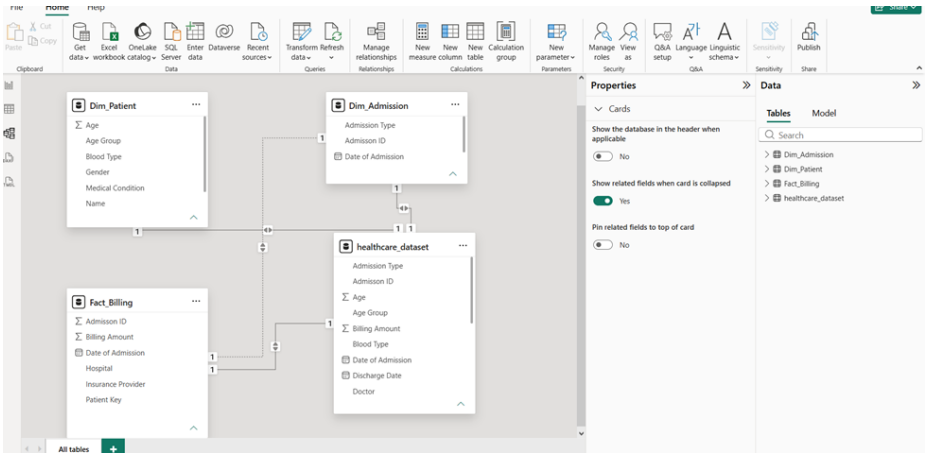
Step 5: Check the tables in Model view

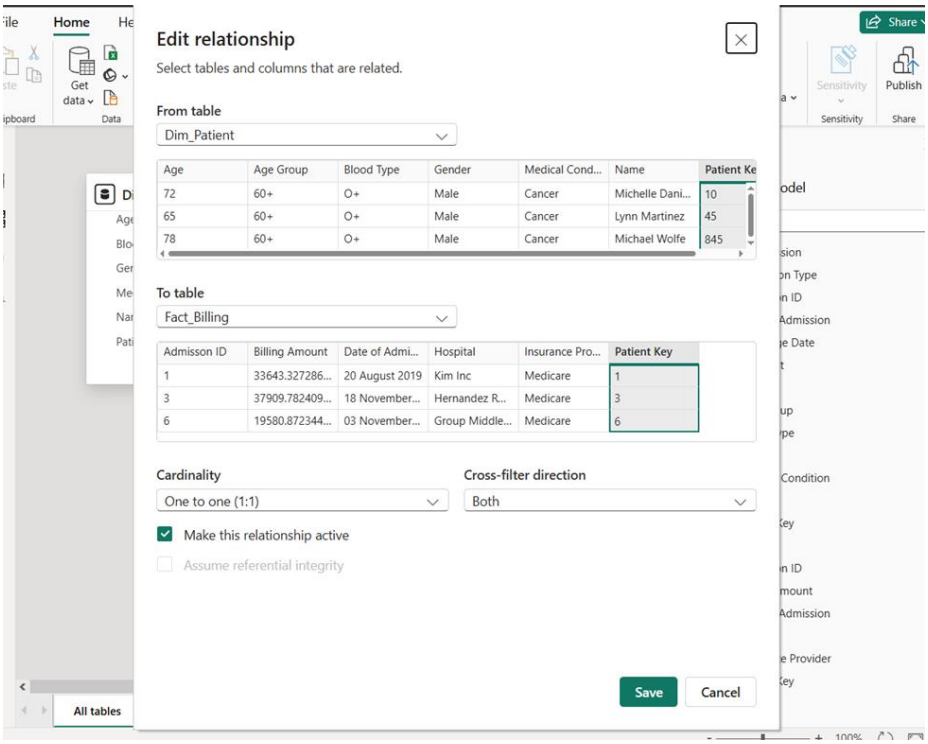
After loading, the Model view showed four tables: Dim_Patient, Dim_Admission, Fact_Billing, and the original healthcare_dataset. At this point none of them were connected to each other yet.



Step 6: Create the relationships

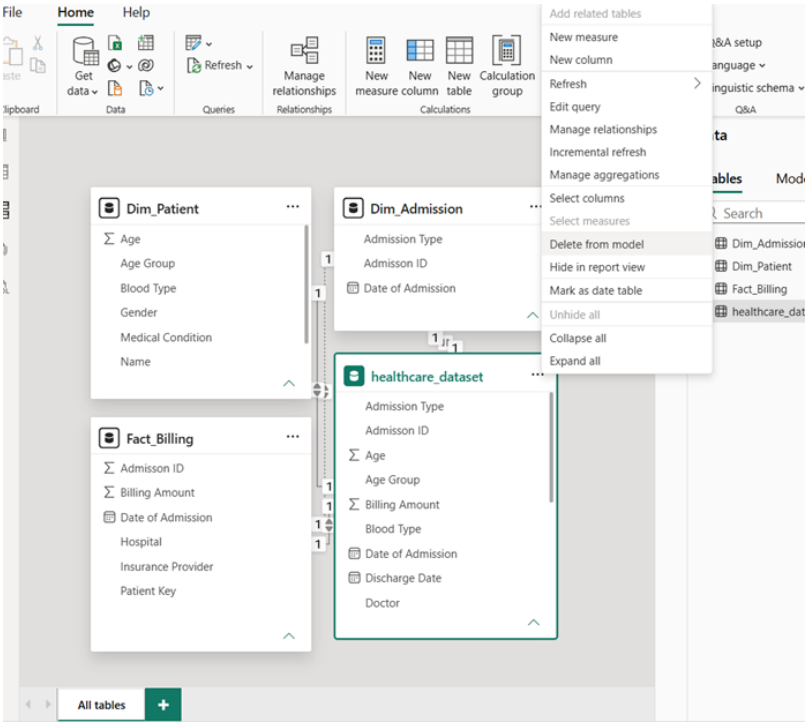
I connected the tables using a Patient Key column: Dim_Patient to Fact_Billing as One to one, with cross-filter direction set to Both and the relationship marked active. I also connected Dim_Admission to the fact and raw tables the same way.





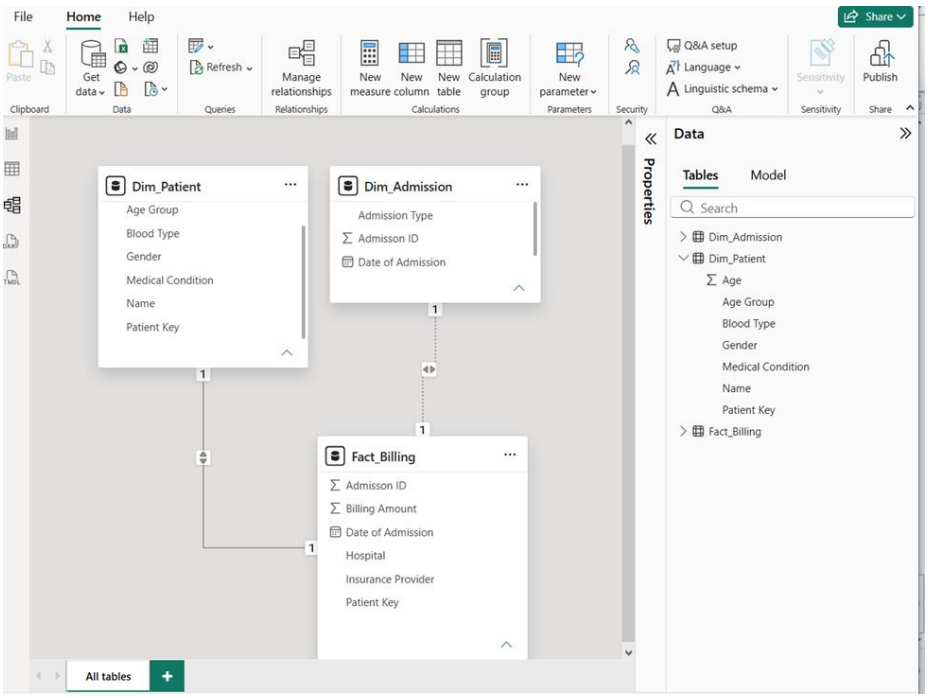
Step 7: Remove the extra raw table

Since I had already split healthcare_dataset into Dim_Patient, Dim_Admission and Fact_Billing, I did not need the original raw table in the model anymore. I right-clicked on healthcare_dataset and chose Delete from model to keep the model clean.



Step 8: Final star-schema model

After removing the raw table, the model was left with just three clean tables: Dim_Patient and Dim_Admission on one side, connected to Fact_Billing in the middle. This is a simple star-schema, with the dimension tables around the fact table.



6. Observations

- The raw data was already fairly clean, so most of the work was in organizing it into separate tables rather than fixing errors.
- Splitting one big table into Dim_Patient, Dim_Admission and Fact_Billing makes the model easier to understand and query.
- The Age Group column makes it much easier to analyze patients by age category instead of using the raw age number.
- Using a Patient Key to join the tables avoids repeating patient details again and again in the fact table.

7. Conclusion

This week I learned how to clean a healthcare dataset in Power Query, create a derived column using conditional logic, and split one big table into proper dimension and fact tables. I also learned how to connect these tables with relationships and remove the extra raw table to keep a clean star-schema model. This kind of model makes it much easier to build an interactive Power BI report for clinical analysis.